

Build a professional React application called:

AI Agent Reliability Cockpit

The application should demonstrate how an AI Agent becomes more reliable over time through:

1. Human review
2. Feedback capture
3. Learned rules
4. Evidence grounding
5. Traceability
6. Edge-case handling
7. Evaluation metrics
8. Governance and human oversight

This should not be a chatbot. It should be an enterprise-grade reliability cockpit that shows how an AI Agent performs a business task, makes an imperfect decision, receives human correction, stores that correction as a reusable rule, and improves its future decision-making.

The objective of the demo is to show:

“AI Agents do not become trustworthy only because the LLM is smarter. They become trustworthy when they are traceable, evidence-grounded, reviewed by humans, measured continuously, and improved using feedback and edge cases.”

Do not mention any specific consulting firm, audit firm, or real company brand anywhere in the application.

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TECH STACK

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Use the following stack:

Frontend:

- React + Vite
- TypeScript
- Tailwind CSS
- shadcn/ui components

- lucide-react icons
- Recharts
- Framer Motion for subtle transitions

Backend:

- Node.js + Express, optional
- In-memory mock data is acceptable
- No external database required
- No LLM API required
- The entire application should work as a deterministic demo

The app should run smoothly on Replit.

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VISUAL DESIGN REQUIREMENTS

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The UI should replicate the look and feel of the uploaded reference screen:

- Left sidebar navigation
- Light background
- Premium consulting-style layout
- Clean card-based UI
- Strong spacing
- Professional typography
- White, slate, blue, indigo, green, amber, and red used carefully
- Minimal but elegant animations
- Executive-ready appearance
- Not cartoonish
- Not hackathon-style
- Not a basic chatbot UI

The sidebar should have:

Logo area:

- AI AGENT
- Reliability Cockpit

Sidebar section label:

- DEMO FLOW

Navigation items:

1. Overview
2. Run Agent
3. Trace Viewer
4. Evidence Map
5. Human Review
6. Learning Log
7. Evaluation Dashboard
8. Business Impact
9. Architecture

Each sidebar item should have:

- Number
- Icon
- Label
- Active state highlight

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CORE DEMO STORY

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The application should tell this story: A business case is submitted to an AI Agent.

The case contains:

- A vendor invoice

- A purchase order
- A policy rule
- Delivery proof status
- Vendor approval status

The AI Agent initially reviews the case and gives an imperfect decision.

The initial decision:

- Risk classification: Medium Risk
- Confidence: 68%
- Reasoning: Vague and incomplete

However, the trace shows that the agent actually detected important issues:

- Invoice amount exceeds purchase order amount
- Delivery proof is missing

A human reviewer then corrects the agent:

“This should be High Risk because the invoice exceeds the PO amount and delivery proof is missing.” The system captures this as a reusable learned rule:

- **If invoice amount exceeds purchase order amount AND delivery proof is missing, classify as High Risk.**

The agent is then rerun on the same case.

After feedback:

- Risk classification changes to High Risk
- Confidence improves to 91%
- Reasoning becomes stronger
- Evidence is explicitly cited
- The learned rule is visible in the Learning Log

The app should clearly show that the agent improved through human feedback and edge-case learning, not model retraining.

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SAMPLE CASE DATA

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Create one primary case:

Case ID:

CASE-2026-001

Case Title:

Vendor Invoice Review

Vendor:

Apex Industrial Services Pvt Ltd

Invoice:

- Invoice ID: INV-7782
- Invoice amount: ₹14,10,000
- Invoice date: 12-Jan-2026
- Vendor status: Approved
- Delivery proof: Missing

Purchase Order:

- PO ID: PO-9912
- Approved PO amount: ₹12,50,000
- PO approval date: 05-Jan-2026
- Approved by: Procurement Manager

Policy Rule:

- If invoice amount exceeds purchase order value by more than ₹50,000, additional review is required.
- If delivery proof is missing, payment should not be approved.
- If invoice exceeds PO amount and delivery proof is missing, the case should be classified as High Risk.

Expected issue:

- Invoice exceeds PO by ₹1,60,000.
- Delivery proof is missing.
- The agent initially underclassifies the case as Medium Risk.
- Human review corrects it to High Risk.
- The learned rule prevents the same mistake in the future.

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EDGE CASE DATA

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Add additional edge cases in the system to show that the AI Agent learns from broader data patterns.

Create a small “Edge Case Library” used by the agent reliability layer:

1. Invoice exceeds PO but delivery proof is available
 - Risk: Medium
 - Reason: Financial mismatch exists, but delivery evidence is present.
2. Invoice is within PO but delivery proof is missing
 - Risk: Medium
 - Reason: Payment should be paused until proof is provided.
3. Invoice exceeds PO and delivery proof is missing
 - Risk: High
 - Reason: Financial mismatch plus missing evidence.
4. Duplicate invoice number found
 - Risk: High
 - Reason: Possible duplicate payment.

5. Vendor bank account differs from master record

- Risk: High
- Reason: Possible fraud or unauthorized vendor update.

6. Invoice date is before PO date

- Risk: Medium
- Reason: Possible process violation or backdated documentation.

7. Emergency approval missing for high-value invoice

- Risk: High
- Reason: Threshold breach without approval.

The application should show that edge cases are used to refine future decisions.

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PAGE 1: OVERVIEW

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Create an Overview page that matches the uploaded UI.

Header:

- SECTION 1 · OPENER
- AI Agent Reliability Cockpit

Subtitle:

Moving beyond chatbots to controlled, traceable digital workers — with visibility, human review, and measurable improvement.

Quote card:

“The secret is not smarter AI. It is better visibility, human review, and measurable improvement. Watch as the same agent gets smarter through feedback — without model retraining or expensive infrastructure.”

Create a section:

THE FOUR PILLARS

Four cards:

1. Traceability

Subtitle: Full audit trail

Description:

Every agent action is logged step-by-step. No black box — users can see exactly what the AI did, in what order, and with what confidence.

2. Evidence Grounding

Subtitle: No hallucinations

Description:

Every claim the agent makes is linked to a source document. Missing evidence is explicitly flagged, not silently ignored.

3. Human Review

Subtitle: Human has final say

Description:

The agent suggests. The human decides. Every high-risk output goes through expert review before action is taken.

4. Continuous Improvement

Subtitle: Gets better over time

Description:

Each human correction becomes a permanent rule. The agent learns from every review without retraining or infrastructure changes.

Create another section:

THE IMPROVEMENT LOOP

Show a horizontal stepper:

- 1. Submit Case
- 2. Agent Suggests
- 3. Human Reviews
- 4. Correction Captured
- 5. Metrics Improve
- 6. Repeat

Each step should have an icon and short description.

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PAGE 2: RUN AGENT

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This page should demonstrate the imperfect first run and improved second run.

Create sections:

1. Case Data

Show a card with:

- Case ID
- Vendor
- Invoice amount
- PO amount
- Difference: ₹1,60,000
- Delivery proof: Missing
- Vendor status: Approved

2. Policy Rule

Show the policy:

- Invoice exceeding PO requires additional review.
- Missing delivery proof requires payment hold.
- Invoice exceeding PO plus missing delivery proof should be High Risk.

3. Agent Output Panel

Before feedback:

- Risk classification: Medium Risk
- Confidence: 68%
- Reason:
“Invoice amount is higher than the purchase order. Additional review may be needed.”

Show this as imperfect because the agent missed the combined severity of:

- Invoice exceeds PO
- Delivery proof missing

Buttons:

- Run Initial Agent
- Rerun After Feedback
- Reset Demo

When clicking “Run Initial Agent”:

- Show a loading state
- Simulate steps
- Display Medium Risk output

When clicking “Rerun After Feedback”:

- Only available after human feedback is submitted
- Display High Risk output

- Confidence should become 91%

- Reason should become:

“Invoice exceeds PO by ₹1,60,000 and delivery proof is missing. This matches the learned high-risk rule and requires human approval before payment.”

Show before and after comparison:\

Before:

- Risk: Medium

- Confidence: 68%

- Rule applied: No

- Evidence completeness: Partial

After:

- Risk: High

- Confidence: 91%

- Rule applied: Yes

- Evidence completeness: Strong

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PAGE 3: TRACE VIEWER

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Create a Trace Viewer page showing every internal step taken by the agent.

This page should prove that the AI Agent is not a black box.

Show a vertical timeline:

Step 1:

Agent: Invoice Extraction Agent

Action: Extract invoice amount

Output: ₹14,10,000

Confidence: 97%

Evidence: INV-7782

Step 2:

Agent: PO Extraction Agent

Action: Extract approved PO amount

Output: ₹12,50,000

Confidence: 96%

Evidence: PO-9912

Step 3:

Agent: Comparison Agent

Action: Compare invoice against PO

Output: Invoice exceeds PO by ₹1,60,000

Confidence: 99%

Evidence: INV-7782 and PO-9912

Step 4:

Agent: Evidence Check Agent

Action: Check delivery proof

Output: Delivery proof missing

Confidence: 94%

Evidence: No delivery proof document found

Step 5:

Agent: Risk Classification Agent

Action: Classify risk

Initial Output: Medium Risk

Confidence: 68%

Issue: Underweighted combined violation

Step 6:

Agent: Human Feedback Agent

Action: Capture reviewer correction

Output: New rule created

Rule: If invoice exceeds PO and delivery proof is missing, classify as High Risk.

Step 7:

Agent: Reliability Layer

Action: Apply learned rule

Output: High Risk

Confidence: 91%

Also add a Trace Details Table with columns:

- Step
- Agent
- Input
- Output
- Confidence
- Evidence ID
- Status
- Notes

Add badges:

- Completed
- Needs Review
- Rule Applied
- Human Corrected

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PAGE 4: EVIDENCE MAP

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Create an Evidence Map page.

This page should show that every AI claim is connected to evidence.

Layout:

Left side:

- List of claims

Right side:

- Evidence details for selected claim

Claims:

1. Invoice amount is ₹14,10,000

Evidence:

- Source: INV-7782
- Field: Invoice total
- Confidence: 97%
- Status: Supported

2. PO amount is ₹12,50,000

Evidence:

- Source: PO-9912
- Field: Approved PO value
- Confidence: 96%
- Status: Supported

3. Invoice exceeds PO by ₹1,60,000

Evidence:

- Source: Derived comparison
- Formula: ₹14,10,000 - ₹12,50,000 = ₹1,60,000
- Confidence: 99%
- Status: Supported

4. Delivery proof is missing

Evidence:

- Source: Document inventory
- Field: Delivery proof file
- Confidence: 94%
- Status: Missing Evidence

5. Vendor is approved

Evidence:

- Source: Vendor master
- Field: Vendor approval status
- Confidence: 92%
- Status: Supported

Create summary cards:

- Supported claims: 4
- Missing evidence items: 1
- Unsupported claims: 0
- Evidence coverage: 80%

Add a message:

“Missing evidence is not treated as a hidden failure. It is surfaced as a reviewable risk signal.”

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PAGE 5: HUMAN REVIEW

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Create a Human Review page.

This is the most important page in the demo.

Show:

1. Agent Decision

- Initial classification: Medium Risk

- Confidence: 68%

- System reasoning:

“Invoice amount is higher than PO amount. Additional review may be needed.”

2. Reviewer Assessment

Create a form with:

- Correct classification dropdown:

- Low Risk

- Medium Risk

- High Risk

- Critical Risk

Default selected:

High Risk

- Reviewer comment:

“This should be High Risk. Invoice exceeds PO by ₹1.6L and delivery proof is missing. Both are control violations.”

- Root cause category:

- Underweighted policy violation

- Missing evidence not escalated

- Weak risk classification

- New edge case

- Other

Default:

Underweighted policy violation

3. Suggested Rule

Show an editable generated rule:

Rule name:

Invoice Over PO + Missing Delivery Proof

Rule logic:

IF invoice_amount > po_amount AND delivery_proof = missing

THEN classify risk as High Risk AND require human approval before payment.

4. Buttons:

- Submit Feedback
- Save as Learned Rule
- Reject Agent Output
- Escalate Case

When clicking Submit Feedback:

- Show success message:

“Feedback captured. New rule added to agent reliability layer.”

- Update Learning Log
- Enable Rerun After Feedback button on Run Agent page
- Update metrics

Add explanation:

“The correction is not just a note. It becomes a reusable rule that improves future decisions.”

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PAGE 6: LEARNING LOG

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Create a Learning Log page.

This page should show how the agent improves over time.

Display entries as cards or a table.

Learning Entry 1:

- Timestamp: current timestamp
- Source: Human Review
- Case ID: CASE-2026-001
- Original decision: Medium Risk
- Corrected decision: High Risk
- Rule created:
If invoice exceeds PO and delivery proof is missing, classify as High Risk.
- Edge case category: Financial mismatch + missing evidence
- Applied to future cases: Yes
- Status: Active

Learning Entry 2:

- Source: Edge Case Library
- Rule:
If duplicate invoice number exists in payment register, classify as High Risk.
- Status: Active

Learning Entry 3:

- Source: Edge Case Library
- Rule:
If vendor bank account does not match vendor master record, classify as High Risk.
- Status: Active

Learning Entry 4:

- Source: Edge Case Library
- Rule:

If invoice date is before PO date, classify as Medium Risk and route for process review.

- Status: Active

Show filters:

- Human feedback
- Edge case
- Policy rule
- Active rules
- Deprecated rules

Add a “Reliability Layer” card:

- Number of learned rules: 4
- Human-generated rules: 1
- Edge-case rules: 3
- Rules applied in latest run: 1

Important explanation:

“This demo does not retrain the LLM. Improvement happens through a reliability layer consisting of human feedback, policy rules, edge-case memory, and deterministic checks.”

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PAGE 7: EVALUATION DASHBOARD

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Create an Evaluation Dashboard page.

This page should show measurable before-and-after improvement.

KPI Cards:

Before Feedback:

- Accuracy: 60%
- Evidence detection: 70%
- Audit trail completeness: 65%

- Missing evidence detection: 55%
- Average confidence: 68%
- Time per case: 35 minutes

After Feedback:

- Accuracy: 88%
- Evidence detection: 95%
- Audit trail completeness: 98%
- Missing evidence detection: 95%
- Average confidence: 91%
- Time per case: 12 minutes

Show delta:

- Accuracy improved by +28 percentage points
- Evidence detection improved by +25 percentage points
- Audit trail completeness improved by +33 percentage points
- Time reduced by 66%

Charts:

1. Before vs After Accuracy
2. Evidence Detection Improvement
3. Audit Trail Completeness
4. Time Per Case Reduction
5. Confidence Improvement

Create a section:

Reliability Metrics

Metrics:

- Rule application rate
- Human override rate

- Unsupported claim rate
- Missing evidence catch rate
- Edge case coverage
- Escalation accuracy
- Reviewer agreement rate

Add explanation:

“Reliability is measured, not assumed. Every feedback event changes future behavior and is reflected in measurable performance metrics.”

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PAGE 8: BUSINESS IMPACT

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Create a Business Impact page. Show why this matters to leadership.

Impact cards:

1. Productivity

- Same team can process up to 3x more cases
- Review time reduced from 35 minutes to 12 minutes

2. Risk Reduction

- More exceptions identified
- Missing evidence surfaced earlier
- Stronger control compliance

3. Governance

- 98% audit trail completeness
- Every decision traceable
- Human review preserved

4. Cost Impact

- Time saved per 1,000 cases:

23,000 minutes

- Hours saved:

383 hours

- Estimated cost saved:

₹19,15,000 per 1,000 cases

- Assumption:

₹5,000 per expert review hour

Create a financial impact table:

Columns:

- Metric
- Manual Process
- AI-Assisted Process
- Improvement

Rows:

- Time per case: 35 min vs 12 min
- Cases per month: 1,000 vs 1,000
- Total hours: 583 vs 200
- Cost: ₹29,15,000 vs ₹10,00,000
- Estimated savings: ₹19,15,000
- Audit trail completeness: 65% vs 98%

Add strategic benefits:

- Faster review cycles
- Better consistency
- Better evidence quality
- Reduced reviewer fatigue

- Lower missed-exception risk
- Stronger governance
- Scalable quality control

Add caveats:

- Human review remains mandatory for high-risk decisions.
- The reliability layer must be governed.
- Learned rules should be versioned and approved.
- Production usage would require integration with document systems, ERP, identity, and access controls.

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PAGE 9: ARCHITECTURE

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Create an Architecture page showing the full operating model.

Show architecture layers:

Layer 1: Case Intake

- Invoice
- PO
- Vendor master
- Policy document
- Delivery evidence

Layer 2: Agent Execution

- Extraction Agent
- Comparison Agent
- Evidence Check Agent
- Risk Classification Agent

Layer 3: Reliability Layer

- Learned rules
- Human feedback
- Edge-case memory
- Confidence thresholds
- Policy checks

Layer 4: Human Review

- Accept
- Reject
- Correct
- Escalate

Layer 5: Governance

- Trace logs
- Evidence map
- Review history
- Rule versioning
- Audit trail

Layer 6: Measurement

- Accuracy
- Evidence coverage
- Override rate
- Confidence
- Time savings
- Business impact

Create a diagram-style layout using cards and arrows.

Add governance principles:

1. The agent suggests; the human decides.

2. Every claim must be evidence-backed.
3. Every correction must be logged.
4. Every learned rule must be versioned.
5. High-risk decisions require human approval.
6. Missing evidence must be surfaced explicitly.
7. Performance must be measured continuously.

Create final statement:

“The goal is not fully autonomous decision-making. The goal is controlled, traceable, auditable, and improving digital work.”

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APPLICATION STATE BEHAVIOR

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The application should maintain demo state:

Initial state:

- No agent run
- No feedback submitted
- No learned rule from human review
- Metrics show baseline

After Run Initial Agent:

- Agent output = Medium Risk
- Confidence = 68%
- Trace populated
- Evidence populated
- Human Review enabled

After Submit Feedback:

- Learned rule created
- Learning Log updated

- Rerun After Feedback enabled
- Metrics partially updated

After Rerun After Feedback:

- Agent output = High Risk
- Confidence = 91%
- Rule applied = Yes
- Evaluation Dashboard shows before/after improvement
- Business Impact shows improved values

After Reset Demo:

- Return to initial state
- Clear learned human feedback
- Reset metrics

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COMPONENT STRUCTURE

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Create this structure:

src/

components/

Sidebar.tsx

Header.tsx

PillarCard.tsx

Stepper.tsx

KPICard.tsx

RiskBadge.tsx

TraceTimeline.tsx

EvidenceCard.tsx

LearningRuleCard.tsx

MetricComparison.tsx

ArchitectureLayer.tsx

pages/

Overview.tsx

RunAgent.tsx

TraceViewer.tsx

EvidenceMap.tsx

HumanReview.tsx

LearningLog.tsx

EvaluationDashboard.tsx

BusinessImpact.tsx

Architecture.tsx

data/

mockCase.ts

mockTrace.ts

mockEvidence.ts

mockRules.ts

mockMetrics.ts

mockEdgeCases.ts

services/

agentSimulator.ts

feedbackService.ts

reliabilityLayer.ts

metricsCalculator.ts

types/

index.ts

App.tsx

main.tsx

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TYPESCRIPT INTERFACES

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Define these interfaces:

```
AuditCase {  
  id: string  
  title: string  
  vendor: string  
  invoiceAmount: number  
  poAmount: number  
  variance: number  
  deliveryProofStatus: "Available" | "Missing"  
  vendorStatus: "Approved" | "Unapproved"  
  policyText: string  
}
```

```
AgentRun {  
  id: string  
  runType: "initial" | "after_feedback"  
  riskClassification: "Low Risk" | "Medium Risk" | "High Risk" | "Critical Risk"  
  confidence: number  
  reasoning: string  
  ruleApplied: boolean  
  timestamp: string  
}
```

```
TraceStep {  
  id: string  
  stepNumber: number  
  agentName: string  
  action: string  
  input: string  
  output: string  
  confidence: number  
  evidenceld?: string  
  status: "Completed" | "Needs Review" | "Rule Applied" | "Human Corrected"  
}
```

```
EvidenceItem {  
  id: string  
  claim: string  
  source: string  
  field: string  
  value: string  
  confidence: number  
  status: "Supported" | "Missing Evidence" | "Unsupported"  
}
```

```
HumanFeedback {  
  id: string  
  caseld: string  
  originalDecision: string  
  correctedDecision: string  
  reviewerComment: string  
  rootCauseCategory: string  
  generatedRule: string  
  timestamp: string
```

}

LearnedRule {

id: string

name: string

source: "Human Feedback" | "Edge Case Library" | "Policy Rule"

condition: string

action: string

status: "Active" | "Inactive"

createdAt: string

}

Metric {

name: string

before: number

after: number

unit: string

}

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IMPORTANT COPY TO INCLUDE IN APP

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Use this copy in the application:

Main headline:

AI Agent Reliability Cockpit

Subtitle:

Moving beyond chatbots to controlled, traceable digital workers — with visibility, human review, and measurable improvement.

Core message:

“The agent is not trusted because it sounds confident. It is trusted because its actions are visible, its claims are evidence-backed, its mistakes are reviewed, and its behavior improves over time.”

Human review message:

“The agent suggests. The human decides. Every correction becomes a reliability asset.”

Learning message:

“This is not model retraining. This is operational learning through rules, feedback, edge cases, and governance.”

Business message:

“The value is not just speed. The value is speed plus reliability, governance, evidence, and measurable improvement.”

Architecture message:

“Every correction becomes an improvement. Every decision is traceable. Every metric is visible.”

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QUALITY REQUIREMENTS

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The final application must:

- Compile without errors
- Have all 9 sidebar pages
- Have working navigation
- Have polished UI
- Have the same demo flow as the reference screen
- Have a working Run Initial Agent button

- Have a working Submit Feedback button
 - Have a working Rerun After Feedback button
 - Have a working Reset Demo button
 - Show before-and-after agent behavior
 - Show traceability
 - Show evidence grounding
 - Show human review
 - Show learning log
 - Show evaluation metrics
 - Show business impact
 - Show architecture
 - Clearly demonstrate that the agent improves based on human review and edge cases
-
- Do not build a chatbot.
 - Do not build a generic dashboard.
 - Do not leave empty placeholder pages.
 - Do not mention any real consulting firm or audit firm.
 - Make it look like an enterprise-grade AI reliability demo suitable for senior business leaders.